

Northwind Retrieval Evaluation Notes

Synthetic research fixture

Purpose	Safe, fictional content for local RAG demonstrations
Topics	Evaluation design, Hybrid retrieval, Context enrichment, Generation checks
Data class	Synthetic / non-confidential

1. Evaluation design

The team evaluates retrieval with a fixed set of forty questions. Each question has chunk-level relevance labels from two reviewers. Disagreements are adjudicated before experiments begin. Recall at ten is the primary retrieval metric.

2. Hybrid retrieval

Dense search retrieves paraphrases well, while lexical search reliably matches identifiers and exact policy terms. Reciprocal rank fusion combines the two ranked lists without requiring their raw scores to share a scale. The constant sixty produced stable results in the pilot.

3. Context enrichment

A short prefix identifies the document, section, and purpose of each chunk. Prefixes are generated during indexing and embedded together with chunk text. Reviewers found that concise prefixes helped ambiguous passages without overwhelming their original wording.

4. Generation checks

Answers are scored for correctness, citation completeness, and abstention when evidence is absent. A second model flags unsupported claims, but human sampling remains necessary because an automated judge can repeat the first model's mistakes.